

Unit 4: Rates, Equilibria and Further Organic Chemistry

IA2 compulsory unit

Externally assessed

Unit description

Introduction

In this unit, students make a quantitative study of chemical kinetics and extend their study of organic reaction mechanisms.

The topics of entropy and equilibria show how chemists are able to predict quantitatively the direction and extent of chemical change.

The unit tests the equilibrium law by showing the degree to which it can accurately predict changes during acid-base reactions, notably the changes to pH during titrations.

The organic chemistry in this unit covers carbonyl compounds, and carboxylic acids and their derivatives.

Students are required to apply their knowledge gained in Units 1 and 2, to all aspects of this unit. This includes nomenclature, ideas of isomerism, bond polarity and bond enthalpy, reagents and reaction conditions, reaction types and mechanisms. Students are also expected to use formulae and balanced equations and calculate chemical quantities.

Chemistry in action

This unit shows how the principles of kinetics and thermodynamics can help to identify optimal conditions for the manufacture of chemicals.

The study of entropy links thermodynamics and equilibrium, and shows how chemists approach fundamental questions about the stability of chemicals and the direction of chemical change.

The historical development of theories explaining acids and bases shows how scientific ideas change as a result of new evidence and fresh thinking.

The study of buffer solutions shows the importance of equilibrium systems in living cells, in medicines, in foods and in the natural environment.

The two broad areas of application of chemistry are synthesis and analysis. In this unit, synthesis is illustrated by reactions of carbonyl compounds (notably with cyanide ions) and the production of esters for use as solvents, flavourings and perfumes. The main analytical technique featured is nuclear magnetic resonance (NMR), including coverage of magnetic resonance imaging.

Practical skills

Through practical work, students will learn about the methods used to measure reaction rates. They will collect data, analyse it and interpret the results. They then see how knowledge of rate equations and other evidence can enable chemists to propose models to describe the mechanisms of reactions.

Simple practical work can be used to investigate equilibrium systems.

Students can develop their skill at volumetric analysis, with a number of titration activities as part of their exploration of acid-base equilibria.

Although the organic section of the unit contains no core practical activities, students would be expected to encounter simple test-tube reactions for organic functional groups – such as the use of Benedict's or Tollens' reagents. There are also opportunities to undertake synthetic reactions, for example to make and purify an ester.

Mathematical skills

There are opportunities for the development of mathematical skills in this unit. This includes plotting and justifying the shapes of rate-concentration and concentration-time graphs, calculating the half-life of a reaction, calculating the activation energy from a suitable graph and rearranging the Arrhenius equation. Also, calculating entropy changes, constructing Born-Haber cycles and calculating missing values, constructing expressions for K_c and K_p and calculating values with relevant units, estimating the change in value of an equilibrium constant when a variable changes, using logarithms and exponentials for converting from concentration to pH of a buffer solution, plotting and interpreting titration curves and representing chiral molecules with appropriate diagrams. There is also an opportunity for calculating R_f values and interpreting infrared spectra and using the $(n + 1)$ rule for proton NMR. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

Assessment information

- First assessment: January 2020.
 - The assessment is 1 hour and 45 minutes.
 - The assessment is out of 90 marks.
 - Students must answer all questions.
 - This paper has three sections:
 - Section A: multiple choice questions
 - Section B: mixture of short-open, open-response, calculations and extended-writing questions
 - Section C: data or calculation question.
 - This paper will contain questions that require information from the Data Booklet (see *Appendix 9*).
 - This paper will include a minimum of 22 marks that target mathematics at Level 2 or above.
 - Students will be expected to apply their knowledge and understanding of experimental methods in familiar and unfamiliar contexts.
 - This paper may contain some synoptic questions which require knowledge and understanding from Units 1 and 2.
 - Calculators may be used in the examination (see *Appendix 8: Use of calculators*).
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Topic 11: Kinetics

Knowledge of the concepts introduced in Unit 2, Topic 9A: Kinetics will be assumed and extended in this topic.

Students will be assessed on their ability to:

11.1	understand the terms: i rate of reaction ii rate equation, $\text{rate} = k[\text{A}]^m[\text{B}]^n$ where m and n are 0, 1 or 2 iii order with respect to a substance in a rate equation iv overall order of a reaction v rate constant vi half-life vii rate-determining step viii activation energy ix heterogeneous and homogeneous catalyst
11.2	be able to calculate the half-life of a reaction, using data from a suitable graph, and identify a reaction with a constant half-life as being first order
11.3	be able to select and justify a suitable experimental technique to obtain rate data for a given reaction, including: i titration ii colorimetry iii mass change iv volume of gas evolved v other suitable technique(s) for a given reaction
11.4	understand experiments that can be used to investigate reaction rates by: i an initial-rate method, carrying out separate experiments where different initial concentrations of one reagent are used <i>A 'clock reaction' is an acceptable approximation of this method.</i> ii a continuous monitoring method to generate data to enable concentration-time or volume-time graphs to be plotted
11.5	be able to deduce the order (0, 1 or 2) with respect to a substance in a rate equation, using data from: i a concentration-time graph ii a rate-concentration graph iii an initial-rate method

11.6	understand how to: - i obtain data to calculate the order with respect to the reactants (and the hydrogen ion) in the acid-catalysed iodination of propanone - ii use these data to make predictions about species involved in the rate-determining step - iii deduce a possible mechanism for the reaction
11.7	be able to deduce the rate-determining step from a rate equation and vice versa
11.8	be able to deduce a reaction mechanism, using knowledge of the rate equation and the stoichiometric equation for a reaction
11.9	understand that knowledge of the rate equations for the hydrolysis of halogenoalkanes can be used to provide evidence for S _N 1 and S _N 2 mechanisms for tertiary and primary halogenoalkane hydrolysis
11.10	be able to use calculations and graphical methods to find the activation energy for a reaction from experimental data <i>The Arrhenius equation will be given if needed.</i>
11.11	understand the use of a solid (heterogeneous) catalyst for industrial reactions, in the gas phase, in terms of providing a surface for the reaction
11.12	CORE PRACTICALS 9a and 9b Following the rate of the iodine-propanone reaction by a titrimetric method and investigating a 'clock reaction' (Harcourt-Esson, iodine clock).
11.13	CORE PRACTICAL 10 Finding the activation energy of a reaction.
	Further suggested practicals: - i the reaction between marble chips and hydrochloric acid (change of mass or change in volume of gas) - ii the reaction between magnesium and hydrochloric acid to determine the activation energy - iii following the rate of the iodine-propanone reaction by a colorimetric method - iv the catalysis by a cobalt(II) salt of potassium sodium tartrate and hydrogen peroxide - v the action of the enzyme urease on urea and thiourea